Q. 1 (Why did the chicken cross the road?)**.** a. Let $\{N_t\}$ be the Poisson process that counts the number of cars. Its rate is $\lambda = 3/10$ on the second timescale. The chicken crosses the road at time $T \sim Exp(\mu)$, where $\mu = 1/10$. We create a Poisson process $\{M_t\}$, independent of $\{N_t\}$, with rate μ that counts the arrivals of chickens, and T is the first arrival time of this process.

Let $R_t = N_t + M_t$. By the superposition principle, this is a Poisson process with rate $\lambda + \mu$, as $\{N_t\}$ and $\{M_t\}$ are independent. It is possible to generate the processes $\{N_t\}$ and $\{M_t\}$ from the superposed process by the thinning principle in the following way. Each arrival of $\{R_t\}$ is independently labeled as a car or a chicken, with probabilities $\frac{\lambda}{\lambda+\mu}$ and $\frac{\mu}{\lambda+\mu}$, respectively; and we can get $\{N_t\}$ as the number of cars and $\{M_t\}$ the number of chickens arriving by time t. Let X_i be the label of the i-th arrival of $\{R_t\}$.

$$
\begin{aligned}
\mathbb{P}(\text{exactly 3 cars pass before the chicken}) &= \mathbb{P}(X_i = \text{ car for } i \in \{1,2,3\}, X_4 = \text{ chicken}) \\
&= \left(\frac{\lambda}{\lambda+\mu}\right)^3 \frac{\mu}{\lambda+\mu} \\
&= \left(\frac{3}{4}\right)^3 \frac{1}{4},
\end{aligned}
$$

the second equality following from independence of the labels.

b. Let T_i denote the time of the i-th arrival of the superposed process $\{R_t\}$. Recall that T is the time at which the first chicken crosses the road. The expected time asked here is

$$
\begin{aligned}
&\mathbb{E}[T | X_i = \text{ car for } i \in \{1, \dots, 5\}, X_6 = \text{ chicken}] \\
&= \mathbb{E}[T_6 | X_i = \text{ car for } i \in \{1, \dots, 5\}, X_6 = \text{ chicken}] \\
&= \mathbb{E}[T_6],
\end{aligned}
$$

as the labels are independent of the arrival times of the superposed process.

And $\mathbb{E}[T_6] = \frac{6}{\lambda+\mu}$, either directly using that $T_6 \sim \Gamma(6, \lambda+\mu)$, or using that T_6 is the sum of six Exponential random variables with rate $\lambda + \mu$ (these Exponentials are even independent but we do not need it for linearity of expectation).

Hence, the desired expected time is

$$\frac{6}{\lambda+\mu} = 15 \text{ seconds.}$$

Grading comments for Q. 1

a. This question was attempted by most students. When the question was correctly set up, most of the students succeeded in finding the right answer. However, some common mistakes are:

   - Not introducing a Poisson process for the arrivals of the chickens and considering T as the first arrival time. You cannot apply superposition to $\{N_t\}$ and the exponential random variable T.
   - Not checking the assumptions required to use the principles of superposition and thinning: namely the independence of the Poisson processes for superposition and the independent labeling of the arrivals for thinning.
   - There is some confusion between the two principles: you obtain the probabilities of the labels from the thinning principle, not the superposition one.
   - Forgetting to justify the last calculation with independence (which comes from the thinning principle).
   - Some students tried to compute this probability directly without using superposition nor thinning but only computed the conditional probability $\mathbb{P}(N_T = 3|T = t)$, instead of $\mathbb{P}(N_T = 3)$ (by tower property). This led to an integral that had to be computed using integration by parts. Partial credit was attributed but are specific to each submission.

   Even if an answer contained the main arguments and obtained the correct result, points were subtracted if assumptions were not checked and some arguments were missing.

b. This question was generally poorly answered. The main contributing factors to this could be:

   - Lack of time: it is apparent that some students attempted this question last and simply ran out of time while still working on it.
   - Misreading/misunderstanding the question and not solving the right problem. Here, it was important to understand that the conditioning event was that the first 5 arrivals were cars and then the chicken crossed. This could be formally seen as $N_T = 5$ (like many of you first approached the problem), but it was best to translate this in terms of labels from the superposed process.
   - Confusion about the arrival times T_k, the counting random variable N_t and the labels, which led to wrong computations.
   - Some students had correctly understood the problem, but mixed up the rates from the Poisson processes $\{N_t\}$ and $\{M_t\}$ with that of $\{R_t\}$, and ended up with the wrong answer but the main argument was correct. Partial credit was of course attributed.

Q. 2 (The easter bunny)**.** a. Let X_i denote the value of the i-th jump. We are given $\mathbb{P}(X_i = -2) = \mathbb{P}(X_i = -1) = \mathbb{P}(X_i = 1) = \mathbb{P}(X_i = 2) = \frac{1}{4}$. Let S_n be the position of the bunny after the n-th jump with initial position $S_0 = 1$. We have

$$S_n = S_0 + \sum_{i=1}^{n} X_i$$

Define the set of values corresponding to the construction holes the bunny can jump into as $F = \{-2, -1, 3, 4\}$. The easter bunny immediately lays one egg at position S_0. The probability the bunny lays one egg at each campus location before he falls into one of the construction holes is then

$$\begin{aligned}
&\mathbb{P}(\{X_1 = -1, X_2 = 2, X_3 \in \{1, 2\}\} \cup \{X_1 = 1, X_2 = -2, X_3 \in \{-1, -2\}\}) \\
&= \mathbb{P}(X_1 = -1, X_2 = 2, X_3 \in \{1, 2\}) + \mathbb{P}(X_1 = 1, X_2 = -2, X_3 \in \{-1, -2\}) \\
&= \mathbb{P}(X_1 = -1)\mathbb{P}(X_2 = 2)\mathbb{P}(X_3 \in \{1, 2\}) + \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = -2)\mathbb{P}(X_3 \in \{-1, -2\}) \\
&= \frac{1}{4} \cdot \frac{1}{4} \cdot \left(\frac{1}{4} + \frac{1}{4}\right) + \frac{1}{4} \cdot \frac{1}{4} \cdot \left(\frac{1}{4} + \frac{1}{4}\right) \\
&= \frac{1}{16},
\end{aligned}$$

where the first equality holds as the events considered are mutually exclusive and the second equality by independence of the jumps.

b. Let T be the number of jumps the bunny makes until he reaches F. We temporarily allow S_0 to take any value, and define $\mu(i) = \mathbb{E}[T|S_0 = i]$. We have $\mu(i) = 0$ for $i \in F$. For $i \notin F$, we use first step analysis to get

$$\mu(i) = 1 + \frac{1}{4}\mu(i-2) + \frac{1}{4}\mu(i-1) + \frac{1}{4}\mu(i+1) + \frac{1}{4}\mu(i+2).$$

Hence, we have

$$\begin{cases}
\mu(0) & = 1 + \frac{1}{4}\mu(1) + \frac{1}{4}\mu(2) \\
\mu(1) & = 1 + \frac{1}{4}\mu(0) + \frac{1}{4}\mu(2) \\
\mu(2) & = 1 + \frac{1}{4}\mu(0) + \frac{1}{4}\mu(1).
\end{cases}$$

Solving this system of 3 equations yields $\mu(0) = \mu(1) = \mu(2) = 2$. The desired expected number of jumps is therefore

$$\mu(1) = 2.$$

Alternative (equally valid) approach:

Instead, we can observe that all the steps have a $1/2$ probability of falling in a construction hole in the immediate next step, irrespective of the particular present state. This is because:

$$\begin{aligned}
\mathbb{P}(S_{n+1} \in F|S_n = 0) &= \mathbb{P}(X_{n+1} \in \{-1, -2\}|S_n = 0) = 1/4 + 1/4 = 1/2 \\
\mathbb{P}(S_{n+1} \in F|S_n = 1) &= \mathbb{P}(X_{n+1} \in \{2, -2\}|S_n = 1) = 1/4 + 1/4 = 1/2 \\
\mathbb{P}(S_{n+1} \in F|S_n = 2) &= \mathbb{P}(X_{n+1} \in \{1, 2\}|S_n = 2) = 1/4 + 1/4 = 1/2.
\end{aligned}$$

And the trials are independent due to the independence of X_i's. So the time when the bunny falls is $T \sim$ Geo$(1/2)$, and $\mathbb{E}[T] = 1/(1/2) = 2$.

Grading comments for Q. 2

a.
- Many of the students got the main idea right. However, a fair few did not formulate the events and/or the random variables denoting states or steps, and relied largely on pictures. These can be helpful, but it is always needed to write down the answers and justifications in words and mathematical terms. Otherwise it is hard to understand what precisely was intended.
- Some tried to use the simple random walk formulae in this setting. That is plainly wrong, since it is not a simple random walk.
- Also, some tried to use first step analysis for the probability calculation. It can address only the "eventual probability", which is not the case here.
- Many of the students correctly tried to enumerate all the paths, but missed a few of them (especially the ones ending with double-length jumps), or thought the fall can happen to only the immediate periphery (states 3, -1).
- Some students simply calculated the probability of one egg appearing at each state after the second jump. This is wrong because it does *not* take into account the event that it must fall immediately after, and also the number of eggs at each state can become more than one.

b.
- For the first approach, some students did not properly define what is here denoted by $\mu(i)$.
- Many students did not (correctly or completely) mention all the boundary conditions. Without them, proceeding to the 3 linear equations in 3 variables are impossible. (Similar to fourth point above)
- Also, some of the equation sets missed the $+1$'s, which was possibly due to step jump. This would make the expected times to be all zero!
- Some calculations fell short due to arithmetic errors.
- Some students directly claimed $\mu(0) = \mu(1) = \mu(2)$ without any explanation, where explanation was required (e.g. permutation symmetry in all three).
- Some students magically produced a solution $(2, 2, 2)$. This is not valid unless it is shown that the linear system is full rank and hence has unique solution (which is the case here).
- Some students tried both methods, but neither completely. In those instances, the better one has been graded.
- For the alternative approach, many student gave no (or incorrect) justification about *why* the probabilities are $1/2$ for each state and the independence of the trials; and in general why the hitting time would be geometric here. Marks have been deducted for these.

Q. 3 (Speculation)**.** a. We have

$$\begin{aligned}\mathbb{P}(S_2 = 2S_1) &= \mathbb{P}(e^{B_2} = 2e^{B_1}) \\ &= \mathbb{P}(B_2 = \ln 2 + B_1) \\ &= \mathbb{P}(B_2 - B_1 = \ln 2) \\ &= 0,\end{aligned}$$

as $B_2 - B_1 \sim \mathcal{N}(0,1)$ is a continuous random variable.

b. The price at time 0 is $S_0 = e^{B_0} = 1$. For the next steps, the justification is written under the computations. We calculate

$$\begin{aligned}\mathbb{P}(S_1 \geq 2, S_2 \leq 1) &= \mathbb{P}(B_1 \geq \ln 2, B_2 \leq 0) \\ &= \mathbb{P}(B_1 \geq \ln 2, B_2 - B_1 \leq -B_1) \\ &= \int_{-\infty}^{\infty} \mathbb{P}(B_1 \geq \ln 2, B_2 - B_1 \leq -x | B_1 = x)\mathbb{P}(B_1 \in dx) \\ &= \int_{\ln 2}^{\infty} \mathbb{P}(B_2 - B_1 \leq -x | B_1 = x)\mathbb{P}(B_1 \in dx) \\ &= \int_{\ln 2}^{\infty} \mathbb{P}(B_2 - B_1 \leq -x)\mathbb{P}(B_1 \in dx) \\ &= \int_{\ln 2}^{\infty} \frac{1}{\sqrt{2\pi}}\Phi(-x)\frac{1}{\sqrt{2\pi}}e^{-x^2/2}dx \\ &= \int_{-\infty}^{-\ln 2} \frac{1}{\sqrt{2\pi}}\Phi(u)\frac{1}{\sqrt{2\pi}}e^{-u^2/2}du \\ &= \frac{1}{2\pi}\frac{1}{2}\Phi(u)^2\Big|_{-\infty}^{-\ln 2} \\ &= \frac{1}{4\pi}\Phi(-\ln 2)^2.\end{aligned}$$

In the above, the third equality comes from the tower property. The fourth one follows from the fact that (i) $B_1 = x$ for $x \geq \ln 2$ obviously implies $B_1 \geq \ln 2$, and (ii) $B_1 = x$ for $x < \ln 2$ is incompatible with $B_1 \geq \ln 2$. Independence of the increments of a Brownian motion gives the fifth equality. The sixth equality uses the CDF (which is $\frac{\Phi}{\sqrt{2\pi}}$ where Φ is defined in the statement) and density formulas for $B_2 - B_1$ and B_1, which both follow $\mathcal{N}(0,1)$ by stationary increments property of the Brownian motion. The change of variable $u = -x$ is used in the seventh equality. We use the hint to obtain the eighth equality.

Grading comments for Q. 3

a.
- Most people are able to notice that the probability is zero since both S_1 and S_2 are continuous. However, simply saying that the probability is zero since the two random variables are continuous is not sufficient justification. We need to manipulate the equality $S_2 = 2S_1$ and reduce it to the equivalent

event $B_2 - B_1 = \ln 2$, which only concerns the probability that a Gaussian distribution is equal to a particular value.

- Some people successfully reduced the probability in question to the probability that $B_2 - B_1 = \ln 2$, but they either failed to identify $B_2 - B_1$ has a Gaussian distribution, or noticed $B_2 - B_1$ is Gaussian but tried to evaluate the probability using the Gaussian density function.
- Another common mistake is saying $\ln(2 \cdot S_1) = \ln(2) \cdot \ln(S_1)$, which should instead be equal to $\ln(2) + \ln(S_1)$.
- A few people stated that S_1 and S_2 are independent, which is not true.
- A few people stated that $\{e^{B_t}\}$ or $e^{B_2 - B_1}$ is a Brownian motion, which is not true.

b.

- Overall part b. is poorly answered. Only a few people made meaningful attempts and wrote something sensible.
- A common mistake is not realizing that since $\{B_t\}$ is a standard Brownian motion, $B_0 = 0$, so S_0 is also a constant. Some people tried to use the tower property and conditioned on the value of S_0, which does not make sense since S_0 is not random.
- Some people had the correct idea of subtracting B_1 from B_2 to use the independent increments property of Brownian motion, but did not know what to do next and did not condition on B_1.
- As in part a., some people said S_1 and S_2 are independent.
- A couple people said $\Phi(x) = 1 - \Phi(-x)$, or said $\int_{-\infty}^{\infty} e^{-x^2/2} dx = 1$, which is not true since the integral is equal to $\sqrt{2\pi}$. Note that this can be derived from the fact that the density of a standard Gaussian integrates to 1.